Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER II
SESSION 2021/2022

COURSE NAME	:	CRYPTOGRAPHY
COURSE CODE	:	BIS20404
PROGRAMME CODE	:	BIS
EXAMINATION DATE	:	JULY 2022
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER **ALL** QUESTIONS. 2. THIS FINAL EXAMINATION IS AN **ONLINE** ASSESSMENT AND CONDUCTED VIA **OPEN** BOOK.

THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

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Q1 (a) Using an example, discuss **ONE (1)** advantage of using modular arithmetic in a cryptosystem.

(4 marks)

(b) Explain why monoalphabetic ciphers are less secured than polyalphabetic ciphers.

(6 marks)

(c) One important property that makes Data Encryption Standard (DES) secure is that the S-boxes are nonlinear. To verify the property, show that $S_1(x_1) \oplus S_1(x_2) \neq S_1(x_1 \oplus x_2)$ for $x_1 = 101010$ and $x_2 = 010101$.

(10 marks)

Q2 (a) Discuss a prominent application of a stream cipher.

(4 marks)

(b) Illustrate the stream cipher in **Q2(a)** using an appropriate diagram.

(6 marks)

(c) Consider a Linear Feedback Shift Register (LFSR) with a degree, $m = 4$, and the feedback path, $p_3 = 0, p_2 = 0, p_1 = 1, p_0 = 1$. If the leftmost bit is computed by XORing two rightmost bits of the feedback path, demonstrate the output of the maximum period generated by the LFSR.

(10 marks)

Q3 (a) Suppose Alice wants to send a message to Bob containing her name N, her computer's IP address, IP, and a request, R to Bob. Assume that Alice and Bob share a symmetric key, K and have securely distributed their public keys, K_S and K_B.

(i) Show a message that enables Bob to verify that the message's integrity has not been violated and that it is from Alice using symmetric key cryptography.

(3 marks)

(ii) Show a message that protects the confidentiality of the request only and ensures that Bob can verify the message's integrity and source using symmetric key cryptography.

(4 marks)

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(iii) Show a message that enables Bob to verify that the message's integrity has not been violated and that it is from Alice using public key cryptography.

(3 marks)

(iv) Show a message that protects the confidentiality of the IP address only and ensures that Bob can verify the message's integrity and source using public key cryptography.

(4 marks)

(b) Discuss the properties of a true cryptographic hash.

(6 marks)

Q4 (a) Compute the value of p, q, and $\phi(n)$ for an $n = 33$ using Rivest-Shamir-Adleman (RSA) cryptosystem.

(5 marks)

(b) Based on **Q4(a),** generate a pair of a public and private key for an $e = 3$. Use Euclidean algorithm to find the inverse modulo.

(10 marks)

(c) Based on **Q4(a),** test any other possible values for e.

(5 marks)

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Q5 (a) A company implements a basic Public Key Infrastructure (PKI) so that the staffs can communicate among them securely. In this case, Bob is hired and then asked to go to Certification Authority (CA) on the first day of his job at the company.

Suppose that Bob already has a pair of public key, pk_B and private key, sk_B while the CA's public and private key are pk_{CA} and sk_{CA} respectively. Apply a digital certificate to protect Bob's public key. Use an appropriate diagram to illustrate the procedure.

(10 marks)

(b) Based on **Q5(a)**, suppose that Bob is communicating with Alice, who is also in the same company. Using appropriate notations, discuss how Alice can verify a digital signature on a message sent by Bob.

(10 marks)

- END OF QUESTIONS –

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